

Modeling the Cart–Pole Inverted Pendulum: Lagrangian Dynamics, Linearization, and LQR Control

Ethan Lemke

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Abstract

This document derives, from first principles, the full nonlinear dynamics of the cart–pole (inverted pendulum) balancer, linearizes them about the upright equilibrium, proves controllability, and derives the LQR balancing controller from the underlying optimal-control theory. It corresponds to Phase 3 (*Understand and model the physics*) and Phase 4 (*Design the balance controller*) of the project roadmap. Phase 5 (energy-based swing-up) and Phase 6 (polish/robustness) are intentionally out of scope here and will be treated in a later document. Numeric physical parameters (pendulum mass/length, cart mass, friction coefficients, travel limits) have not yet been measured on the hardware; every result below is therefore left symbolic so that measured values can simply be substituted in once available.

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Part I

Physics Modeling (Phase 3)

1 System Description and Notation

The physical system is a motorized carriage that translates horizontally along a linear rail (belt-driven by a NEMA17 stepper through a TMC2209 driver), with a rigid steel rod pendulum pivoted to the carriage through an LPD3806 rotary encoder. The encoder measures the pendulum's angle relative to the carriage; the stepper effectively lets us command a horizontal force (or, at the step-command level, a position/velocity) on the carriage.

1.1 Modeling assumptions

- Planar motion only (no out-of-rail-plane dynamics).
- The rail is horizontal and rigid; the carriage moves without vertical motion.
- The pendulum is a *rigid, uniform rod* — not a point mass. Its rotational inertia about its own center of mass matters and is carried through the derivation explicitly (this is what makes the model a *physical* pendulum rather than the simplified *simple* pendulum many textbook treatments use).
- Belt/drivetrain dynamics (stepper torque ripple, belt compliance, backlash, discretized step motion) are abstracted away: we model the actuator as producing an ideal continuous horizontal force u directly on the carriage. This is a real modeling gap to revisit during Phase 6 (robustness) once the idealized controller is working; it is standard practice to idealize the actuator first and

treat its non-idealities as disturbances/uncertainty to be handled by controller robustness margin and empirical tuning, rather than modeled analytically up front.

- Friction is modeled as simple viscous (linear-in-velocity) damping at the carriage/rail interface and at the pivot bearing. Coulomb (static/kinetic) friction and any pivot backlash are neglected for now.

1.2 Coordinates and parameters

Symbol	Meaning
x	Carriage (cart) position along the rail [m]
θ	Pendulum angle, measured from the <i>upward</i> vertical, positive in the $+x$ tilt direction [rad]
m_c	Carriage mass [kg]
m_p	Pendulum (rod) mass [kg]
L	Full rod length [m]
$\ell_c = L/2$	Distance from pivot to rod center of mass (uniform rod) [m]
$I_{\text{cm}} = \frac{1}{12}m_p L^2$	Rod's moment of inertia about its own center of mass
$I_p = I_{\text{cm}} + m_p \ell_c^2 = \frac{1}{3}m_p L^2$	Rod's moment of inertia about the pivot (see §1.2)
g	Gravitational acceleration [m/s ²]
b_x	Viscous friction coefficient, carriage/rail [N·s/m]
b_θ	Viscous friction coefficient, pivot bearing [N·m·s/rad]
u	Control input: horizontal force applied to the carriage [N]

Notation collision, flagged up front: controls literature universally writes the *state vector* in bold as $\mathbf{x} = [x, \dot{x}, \theta, \dot{\theta}]^\top$, while the very same letter x (not bold) is the cart's position — one of the four entries of $\mathbf{x}$. Both conventions are too standard to abandon, so both are kept; bold vs. not bold disambiguates throughout.

Figure 1 shows the sign convention: $\theta = 0$ is straight up (the unstable equilibrium we want to balance at), and positive θ tilts the rod toward $+x$.

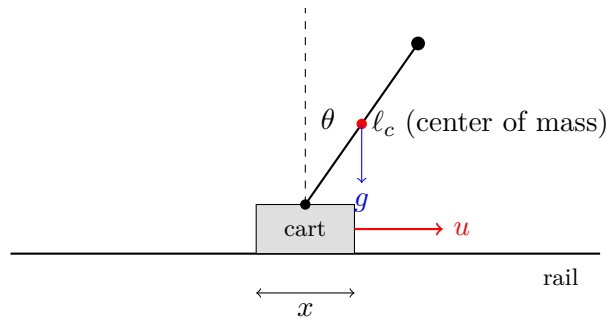


Figure 1: Cart–pole sign convention. θ measured from upward vertical; u is the horizontal actuator force on the cart.

Tangent — the parallel axis theorem.

For a rigid body of mass m , the moment of inertia about any axis parallel to one through the center of mass is

$$I_{\text{axis}} = I_{\text{cm}} + md^2,$$

where d is the distance between the two parallel axes. This follows directly from expanding $I_{axis} = \int \|\mathbf{r} - \mathbf{r}_{axis}\|^2 dm$ in terms of the center-of-mass position and using $\int (\mathbf{r} - \mathbf{r}_{cm}) dm = 0$ by definition of the center of mass — the cross term vanishes, leaving exactly $I_{cm} + md^2$. For a uniform rod of length L pivoted at one end, $d = \ell_c = L/2$ and $I_{cm} = \frac{1}{12}mL^2$, giving the familiar $I_p = \frac{1}{12}mL^2 + m(L/2)^2 = \frac{1}{3}mL^2$.

2 Lagrangian Mechanics Primer

Tangent — why Lagrangian mechanics instead of Newton–Euler here.

Newton–Euler mechanics requires drawing every free body (cart, rod), introducing the unknown constraint/reaction forces at the pivot and rail (normal forces, pivot pin reactions), writing $F = ma$ and $\tau = I\alpha$ for each body, and then algebraically eliminating the reaction forces to get back down to two equations in x, θ . Lagrangian mechanics reaches the same two equations directly, because the reaction/constraint forces of an ideal constraint (rigid rod, frictionless-in-the-constraint- direction rail) do zero virtual work and simply never appear. For a two-body, two-constrained-DOF system like this one, that is a substantial simplification.

Tangent — generalized coordinates, Hamilton’s principle, and the Euler–Lagrange equation.

A generalized coordinate set $q_1, \dots, q_n$ is any minimal set of parameters that fully determines the configuration of the system — here $n = 2$, $q = (x, \theta)$, since fixing x and θ fixes the position of every point of the cart and rod. Define the Lagrangian $\mathcal{L}(q, \dot{q}) = T(q, \dot{q}) - V(q)$, kinetic minus potential energy, and the action functional over a candidate trajectory $q(t)$ on $[t_0, t_1]$ with fixed endpoints:

$$S[q] = \int_{t_0}^{t_1} \mathcal{L}(q(t), \dot{q}(t)) dt.$$

Hamilton’s principle states that the true trajectory of the system is a stationary point of S : $\delta S = 0$ for all admissible variations $\delta q(t)$ vanishing at the endpoints. Writing $q \rightarrow q + \epsilon \eta$ for a smooth perturbation η with $\eta(t_0) = \eta(t_1) = 0$, and differentiating S with respect to ϵ at $\epsilon = 0$:

$$0 = \left. \frac{d}{d\epsilon} \right|_0 S[q + \epsilon \eta] = \int_{t_0}^{t_1} \left(\frac{\partial \mathcal{L}}{\partial q} \eta + \frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{\eta} \right) dt.$$

Integrating the second term by parts (the boundary term vanishes because $\eta(t_0) = \eta(t_1) = 0$) gives $\int \left(\frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} \right) \eta dt = 0$ for every admissible η , so by the fundamental lemma of the calculus of variations the bracketed quantity must vanish identically:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = 0, \quad i = 1, \dots, n.$$

This is the Euler–Lagrange equation. When non-conservative (externally applied or dissipative) generalized forces Q_i act on coordinate q_i — here, the motor force and friction torque/force, which

are not derivable from a potential — the same derivation with virtual work $\sum_i Q_i \delta q_i$ added to the variation yields the more general form actually used below:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i.$$

3 Kinematics

Take the pivot (top of the carriage) as the reference point. With θ measured from upward vertical as in Figure 1, the pivot sits at $(x, 0)$ and the rod's center of mass sits at

$$\mathbf{r}_{\text{cm}} = (x + \ell_c \sin \theta, \ell_c \cos \theta).$$

Differentiating,

$$\dot{\mathbf{r}}_{\text{cm}} = (\dot{x} + \ell_c \cos \theta \dot{\theta}, -\ell_c \sin \theta \dot{\theta}),$$

so the squared speed of the center of mass is

$$\begin{aligned} v_{\text{cm}}^2 &= (\dot{x} + \ell_c \cos \theta \dot{\theta})^2 + (\ell_c \sin \theta \dot{\theta})^2 \\ &= \dot{x}^2 + 2\ell_c \cos \theta \dot{x} \dot{\theta} + \ell_c^2 \cos^2 \theta \dot{\theta}^2 + \ell_c^2 \sin^2 \theta \dot{\theta}^2 \\ &= \dot{x}^2 + 2\ell_c \cos \theta \dot{x} \dot{\theta} + \ell_c^2 \dot{\theta}^2, \end{aligned}$$

using $\sin^2 \theta + \cos^2 \theta = 1$.

4 Energy and the Lagrangian

4.1 Kinetic energy

The carriage translates without rotating: $T_{\text{cart}} = \frac{1}{2} m_c \dot{x}^2$. The rod has both translational kinetic energy of its center of mass and rotational kinetic energy about that center of mass:

$$T_{\text{rod}} = \frac{1}{2} m_p v_{\text{cm}}^2 + \frac{1}{2} I_{\text{cm}} \dot{\theta}^2.$$

Summing and substituting v_{cm}^2 from above,

$$\begin{aligned} T &= \frac{1}{2} m_c \dot{x}^2 + \frac{1}{2} m_p \left(\dot{x}^2 + 2\ell_c \cos \theta \dot{x} \dot{\theta} + \ell_c^2 \dot{\theta}^2 \right) + \frac{1}{2} I_{\text{cm}} \dot{\theta}^2 \\ &= \frac{1}{2} (m_c + m_p) \dot{x}^2 + m_p \ell_c \cos \theta \dot{x} \dot{\theta} + \frac{1}{2} (I_{\text{cm}} + m_p \ell_c^2) \dot{\theta}^2 \\ &= \frac{1}{2} (m_c + m_p) \dot{x}^2 + m_p \ell_c \cos \theta \dot{x} \dot{\theta} + \frac{1}{2} I_p \dot{\theta}^2, \end{aligned} \tag{1}$$

where the last step used $I_p = I_{\text{cm}} + m_p \ell_c^2$ from the parallel axis theorem (§1.2) to fold the translational and rotational rod terms into a single moment of inertia about the pivot.

4.2 Potential energy

Taking the pivot height as the zero reference,

$$V = m_p g \ell_c \cos \theta. \quad (2)$$

Note V is *maximized* at $\theta = 0$ (upright) and minimized at $\theta = \pi$ (hanging straight down) — consistent with upright being the unstable, high-potential-energy equilibrium we are trying to balance at, and hanging-down being the passive, stable, low-energy equilibrium.

Tangent — total mechanical energy and power balance.

Define $E = T + V$. Differentiating along any trajectory and using the equations of motion derived in the next section, one finds $\dot{E} = u\dot{x} - b_x\dot{x}^2 - b_\theta\dot{\theta}^2$: the actuator injects power $u\dot{x}$, and friction dissipates power $b_x\dot{x}^2 + b_\theta\dot{\theta}^2 \geq 0$. With $u = 0$ and no friction, E is exactly conserved — a good numerical sanity check for a simulator (integrate the frictionless, unforced equations and confirm E stays constant to integrator tolerance). This E is also the quantity a swing-up controller (Phase 5, deferred) would regulate toward the upright value $E^* = m_p g \ell_c$ — noted here for completeness since it falls straight out of T and V , not because swing-up is being designed now.

4.3 The Lagrangian

$$\mathcal{L} = T - V = \frac{1}{2}(m_c + m_p)\dot{x}^2 + m_p \ell_c \cos \theta \dot{x} \dot{\theta} + \frac{1}{2}I_p \dot{\theta}^2 - m_p g \ell_c \cos \theta. \quad (3)$$

5 Nonlinear Equations of Motion

The non-conservative generalized forces are the actuator force (acting on x) and viscous friction (acting on both coordinates, opposing motion):

$$Q_x = u - b_x \dot{x}, \quad Q_\theta = -b_\theta \dot{\theta}.$$

5.1 The x equation

From (3), $\frac{\partial \mathcal{L}}{\partial \dot{x}} = (m_c + m_p)\dot{x} + m_p \ell_c \cos \theta \dot{\theta}$ and $\frac{\partial \mathcal{L}}{\partial x} = 0$ (x does not appear explicitly in $\mathcal{L}$ — see the cyclic-coordinate remark in §6). So

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} = (m_c + m_p)\ddot{x} + m_p \ell_c \cos \theta \ddot{\theta} - m_p \ell_c \sin \theta \dot{\theta}^2,$$

and the Euler–Lagrange equation $\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} - \frac{\partial \mathcal{L}}{\partial x} = Q_x$ gives

$$(m_c + m_p)\ddot{x} + m_p \ell_c \cos \theta \ddot{\theta} - m_p \ell_c \sin \theta \dot{\theta}^2 + b_x \dot{x} = u. \quad (4)$$

5.2 The θ equation

$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m_p \ell_c \cos \theta \dot{x} + I_p \dot{\theta}$, so

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m_p \ell_c \cos \theta \ddot{x} - m_p \ell_c \sin \theta \dot{x} \dot{\theta} + I_p \ddot{\theta}.$$

Also $\frac{\partial \mathcal{L}}{\partial \theta} = -m_p \ell_c \sin \theta \dot{x} + m_p g \ell_c \sin \theta$ (differentiating the $\cos \theta$ terms in $\mathcal{L}$). Subtracting,

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} - \frac{\partial \mathcal{L}}{\partial \theta} = m_p \ell_c \cos \theta \ddot{x} + I_p \ddot{\theta} - m_p g \ell_c \sin \theta,$$

the $-m_p \ell_c \sin \theta \dot{x} \dot{\theta}$ Coriolis-type term canceling exactly between the two pieces. Setting this equal to Q_θ :

$$I_p \ddot{\theta} + m_p \ell_c \cos \theta \ddot{x} - m_p g \ell_c \sin \theta + b_\theta \dot{\theta} = 0. \quad (5)$$

Equations (4)–(5) are the full nonlinear cart-pole dynamics — a coupled pair of second-order ODEs in x, θ , exact (given the stated modeling assumptions), for any amplitude of θ .

5.3 Manipulator (mass-matrix) form

It is standard, and useful for simulation, to write (4)–(5) as $M(\theta)\ddot{\mathbf{q}} + \mathbf{c}(\theta, \dot{\theta}) + \mathbf{d}(\dot{\mathbf{q}}) + \mathbf{g}(\theta) = \mathbf{B}u$, with $\mathbf{q} = (x, \theta)^\top$:

$$M(\theta) = \begin{bmatrix} m_c + m_p & m_p \ell_c \cos \theta \\ m_p \ell_c \cos \theta & I_p \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} -m_p \ell_c \sin \theta \dot{\theta}^2 \\ 0 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} b_x \dot{x} \\ b_\theta \dot{\theta} \end{bmatrix},$$

$$\mathbf{g}(\theta) = \begin{bmatrix} 0 \\ -m_p g \ell_c \sin \theta \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

$M(\theta)$ is symmetric positive definite for all θ (it is a kinetic-energy/mass matrix — physically, kinetic energy $T = \frac{1}{2} \dot{\mathbf{q}}^\top M \dot{\mathbf{q}}$ must be positive for any nonzero velocity), so it is always invertible; this matters in the next step.

5.4 Explicit form for numerical simulation

To simulate, solve the 2×2 linear system $M(\theta)\ddot{\mathbf{q}} = \mathbf{B}u - \mathbf{c} - \mathbf{d} - \mathbf{g}$ for $\ddot{x}, \ddot{\theta}$ explicitly. Define

$$D(\theta) = \det M(\theta) = (m_c + m_p)I_p - m_p^2 \ell_c^2 \cos^2 \theta > 0, \quad (6)$$

(strictly positive by the positive-definiteness of M just noted). Then

$$\ddot{x} = \frac{I_p [u - b_x \dot{x} + m_p \ell_c \sin \theta \dot{\theta}^2] - m_p \ell_c \cos \theta [m_p g \ell_c \sin \theta - b_\theta \dot{\theta}]}{D(\theta)}, \quad (7)$$

$$\ddot{\theta} = \frac{(m_c + m_p) [m_p g \ell_c \sin \theta - b_\theta \dot{\theta}] - m_p \ell_c \cos \theta [u - b_x \dot{x} + m_p \ell_c \sin \theta \dot{\theta}^2]}{D(\theta)}. \quad (8)$$

Together with $\dot{x} = \dot{x}$ and $\dot{\theta} = \dot{\theta}$ (trivially), (7)–(8) give four coupled first-order ODEs $\dot{\mathbf{x}} = f(\mathbf{x}, u)$ in the state $\mathbf{x} = (x, \dot{x}, \theta, \dot{\theta})^\top$ — this is exactly what gets coded up (e.g. in Python with `scipy.integrate.solve_ivp`, or a hand-rolled RK4 step) to simulate the nonlinear pendulum.

Tangent — sanity-checking a nonlinear simulation before trusting it.

Two cheap checks catch most implementation bugs. (1) Energy conservation: with $u = 0$, $b_x = b_\theta = 0$, and any initial condition, $E = T + V$ computed from the simulated trajectory should stay constant to numerical-integrator tolerance (see the energy tangent above) — if it drifts, there's a sign or algebra error. (2) Small-oscillation check: released from rest near $\theta = \pi$ (hanging down)

with x held fixed conceptually, the linearized period should match the classical physical-pendulum result $T_{osc} = 2\pi\sqrt{I_p/(m_p g \ell_c)}$ — if the simulated small-swing period near the downward equilibrium doesn't match that closed form, something upstream is wrong.

Part II

Linearization and Controllability (Phase 3, continued)

6 Equilibria and Linearization

6.1 Equilibria

An equilibrium of $\dot{\mathbf{x}} = f(\mathbf{x}, u)$ with $u = 0$ requires $\dot{x} = \dot{\theta} = 0$ (so $\ddot{x} = \ddot{\theta} = 0$ too) and, from (7)–(8) with all velocity terms and u zero, $\sin\theta = 0$, i.e. $\theta \in \{0, \pi\}$: upright and hanging-down. Notice x itself never appears on the right-hand side of (7)–(8) — *any* value of x is equally an equilibrium. This document only needs $\theta = 0$ (upright), since that is the equilibrium LQR will stabilize; $\theta = \pi$ matters for Phase 5 (swing-up start condition), deferred.

Tangent — why x drops out entirely: a cyclic coordinate.

$\mathcal{L}$ in (3) does not depend on x itself, only on $\dot{x}$ — x is called a *cyclic* (or *ignorable*) coordinate. Physically this is a translation symmetry: sliding the whole cart-pole system sideways changes nothing about its dynamics. For a cyclic coordinate in an unforced, frictionless Lagrangian system, the associated generalized momentum $p_x = \partial\mathcal{L}/\partial\dot{x}$ is conserved (Noether's theorem, translation symmetry $\rightarrow$ momentum conservation) — here, with the applied force u and friction $b_x\dot{x}$ present, p_x is not literally conserved, but the structural fact driving both statements is the same: $\partial\mathcal{L}/\partial x \equiv 0$. That is exactly why x never appears on the right of (4)–(5), and hence why any x is an equilibrium value — the physically sensible statement, since the cart-pole should balance the same way no matter where along the rail it happens to be (travel limits aside).

6.2 Linearizing about the upright equilibrium

Tangent — Taylor-expansion linearization in general.

For $\dot{\mathbf{x}} = f(\mathbf{x}, u)$ with equilibrium $f(\mathbf{x}^*, u^*) = 0$, write $\mathbf{x} = \mathbf{x}^* + \delta\mathbf{x}$, $u = u^* + \delta u$, and expand f to first order:

$$\dot{\delta\mathbf{x}} \approx \underbrace{\left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\star}}_A \delta\mathbf{x} + \underbrace{\left. \frac{\partial f}{\partial u} \right|_{\star}}_B \delta u,$$

dropping $O(\|\delta\mathbf{x}\|^2, \|\delta u\|^2)$ terms — this is only accurate in a neighborhood of $(\mathbf{x}^*, u^*)$; for a pendulum the practically useful radius is roughly $\pm 15^\circ$ – 20° before the small-angle approximation visibly degrades, though the exact acceptable range depends on how aggressively the closed loop drives θ .

At $\theta^* = 0$, $\dot{x}^* = \dot{\theta}^* = 0$, $u^* = 0$: apply $\sin \theta \approx \theta$, $\cos \theta \approx 1$, and drop every term that is already a product of two small quantities (in particular $\sin \theta \dot{\theta}^2 \approx \theta \dot{\theta}^2$ is third-order small and vanishes entirely at linear order — it does not survive as $\theta \dot{\theta}^2$; it is dropped, full stop). Equations (4)–(5) become linear:

$$(m_c + m_p)\ddot{x} + m_p \ell_c \ddot{\theta} + b_x \dot{x} = u, \quad (9)$$

$$I_p \ddot{\theta} + m_p \ell_c \ddot{x} - m_p g \ell_c \theta + b_\theta \dot{\theta} = 0. \quad (10)$$

Writing $M_0 = \begin{bmatrix} m_c + m_p & m_p \ell_c \\ m_p \ell_c & I_p \end{bmatrix}$ (i.e. $M(\theta)$ at $\theta = 0$) and $D_0 = \det M_0 = (m_c + m_p)I_p - m_p^2 \ell_c^2 =$

$D(0)$ from (6), solving (9)–(10) for $\ddot{x}, \ddot{\theta}$ (via $M_0^{-1} = \frac{1}{D_0} \begin{bmatrix} I_p & -m_p \ell_c \\ -m_p \ell_c & m_c + m_p \end{bmatrix}$) gives

$$\ddot{x} = \frac{1}{D_0} \left[I_p u - I_p b_x \dot{x} - m_p^2 g \ell_c^2 \theta + m_p \ell_c b_\theta \dot{\theta} \right], \quad (11)$$

$$\ddot{\theta} = \frac{1}{D_0} \left[-m_p \ell_c u + m_p \ell_c b_x \dot{x} + (m_c + m_p) m_p g \ell_c \theta - (m_c + m_p) b_\theta \dot{\theta} \right]. \quad (12)$$

6.3 State-space matrices

With $\mathbf{x} = (x, \dot{x}, \theta, \dot{\theta})^\top$, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{I_p b_x}{D_0} & -\frac{m_p^2 g \ell_c^2}{D_0} & \frac{m_p \ell_c b_\theta}{D_0} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{m_p \ell_c b_x}{D_0} & \frac{(m_c + m_p) m_p g \ell_c}{D_0} & -\frac{(m_c + m_p) b_\theta}{D_0} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{I_p}{D_0} \\ 0 \\ -\frac{m_p \ell_c}{D_0} \end{bmatrix}. \quad (13)$$

This is the linear model LQR (Part II) is designed against. Its first and third rows are the trivial kinematic identities $\dot{x} = \dot{x}$, $\dot{\theta} = \dot{\theta}$; all the physics is in rows two and four, read directly off (11)–(12).

7 Controllability

Tangent — what controllability means, and the Kalman rank test.

A linear system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$ ($\mathbf{x} \in \mathbb{R}^n$) is controllable if, for any initial state and any target state, there exists a finite-time input trajectory $u(t)$ driving one to the other. Kalman's rank test says this holds iff the controllability matrix

$$\mathcal{C} = [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}]$$

has full row rank n . Why this is the right test: the set of states reachable from the origin using inputs on $[0, t]$ is (a closure of) the span of $\{\mathbf{A}^k \mathbf{B}\}_{k \geq 0}$ contracted through the variation-of-constants formula $\mathbf{x}(t) = \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B}u(\tau) d\tau$, and by the Cayley–Hamilton theorem every power $\mathbf{A}^k$ for $k \geq n$ is expressible as a linear combination of $I, \mathbf{A}, \dots, \mathbf{A}^{n-1}$ (since $\mathbf{A}$ satisfies its own degree- n characteristic polynomial) — so the reachable subspace is exactly spanned by the first n terms $\mathbf{B}, \mathbf{A}\mathbf{B}, \dots, \mathbf{A}^{n-1}\mathbf{B}$, no more information is added by going further. (An equivalent, often more diagnostic, alternative is

the Popov–Belevitch–Hautus test: (A, B) is controllable iff $\text{rank}[A - \lambda I \ B] = n$ for every eigenvalue λ of A — i.e. no eigenvector of $A^\top$ is orthogonal to every column of B , meaning no mode of the system is invisible to the actuator.)

Here $n = 4$, so we need $\text{rank } \mathcal{C} = 4$ for A, B from (13).

7.1 Exact symbolic proof (frictionless case)

The friction terms complicate the algebra without changing the qualitative answer, so first set $b_x = b_\theta = 0$ to see the structure cleanly (the frictional case is addressed just after). Write

$$\alpha = -\frac{m_p^2 g \ell_c^2}{D_0}, \quad \beta = \frac{(m_c + m_p) m_p g \ell_c}{D_0}, \quad b_2 = \frac{I_p}{D_0}, \quad b_4 = -\frac{m_p \ell_c}{D_0},$$

so that (with $b_x = b_\theta = 0$)

$$A_0 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \alpha & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \beta & 0 \end{bmatrix}, \quad B_0 = \begin{bmatrix} 0 \\ b_2 \\ 0 \\ b_4 \end{bmatrix}.$$

Multiplying out successively (each application of A_0 just shifts and reweights entries, since A_0 is so sparse):

$$A_0 B_0 = \begin{bmatrix} b_2 \\ 0 \\ b_4 \\ 0 \end{bmatrix}, \quad A_0^2 B_0 = \begin{bmatrix} 0 \\ \alpha b_4 \\ 0 \\ \beta b_4 \end{bmatrix}, \quad A_0^3 B_0 = \begin{bmatrix} \alpha b_4 \\ 0 \\ \beta b_4 \\ 0 \end{bmatrix}.$$

So

$$\mathcal{C} = \begin{bmatrix} 0 & b_2 & 0 & \alpha b_4 \\ b_2 & 0 & \alpha b_4 & 0 \\ 0 & b_4 & 0 & \beta b_4 \\ b_4 & 0 & \beta b_4 & 0 \end{bmatrix}.$$

$\mathcal{C}$ has a checkerboard sparsity pattern — columns 1, 3 are nonzero only in rows 2, 4, and columns 2, 4 only in rows 1, 3. Permuting rows to (2, 4, 1, 3) and columns to (1, 3, 2, 4) block-diagonalizes it into two copies of the same 2×2 block $\begin{bmatrix} b_2 & \alpha b_4 \\ b_4 & \beta b_4 \end{bmatrix}$, so (a permutation only changes the sign of a determinant, never whether it's zero)

$$\det \mathcal{C} = \pm [b_2 \beta b_4 - \alpha b_4^2]^2 = \pm b_4^2 (b_2 \beta - \alpha b_4)^2.$$

Substituting the definitions of α, β, b_2, b_4 :

$$b_2 \beta - \alpha b_4 = \frac{I_p (m_c + m_p) m_p g \ell_c}{D_0^2} - \frac{m_p^3 g \ell_c^3}{D_0^2} = \frac{m_p g \ell_c [I_p (m_c + m_p) - m_p^2 \ell_c^2]}{D_0^2} = \frac{m_p g \ell_c D_0}{D_0^2} = \frac{m_p g \ell_c}{D_0},$$

using the definition $D_0 = (m_c + m_p) I_p - m_p^2 \ell_c^2$ exactly cancels the bracket. Therefore

$$\det \mathcal{C} = b_4^2 \left(\frac{m_p g \ell_c}{D_0} \right)^2 = \left(\frac{m_p \ell_c}{D_0} \right)^2 \left(\frac{m_p g \ell_c}{D_0} \right)^2 = \frac{m_p^4 g^2 \ell_c^4}{D_0^4}. \quad (14)$$

Since $m_p > 0$, $g > 0$, $\ell_c > 0$ (any real rod) and $D_0 > 0$ (positive definiteness of M_0 , established earlier), (14) is *strictly positive for every physically valid parameter choice* — there is no degenerate case to worry about (short of $m_p = 0$ or $\ell_c = 0$, i.e., no pendulum at all). This is a clean, exact result: **the frictionless linearized cart-pole is always controllable.**

Tangent — the physical intuition behind this result.

Controllability here hinges entirely on $m_p \ell_c \neq 0$ appearing in the off-diagonal coupling of M_0 — physically, that off-diagonal term is what lets a horizontal push on the cart exert a torque on the pendulum about the pivot (equivalently, by Newton’s third law, what lets the pendulum’s motion react back onto the cart). If the pendulum had zero mass or zero length, that coupling vanishes and the actuator could no longer influence θ at all — hence $m_p \ell_c$ appearing, squared, directly in (14).

7.2 Effect of friction

With $b_x, b_\theta > 0$, A gains nonzero $(2, 2)$ and $(4, 4)$ diagonal entries and $\mathcal{C}$ loses the exact checkerboard symmetry used above, so the determinant no longer collapses into the clean closed form (14). Controllability is a *generic* property (the set of (A, B) pairs that fail the rank test is measure-zero in parameter space), and friction enters as a continuous perturbation of A , so the physically expected outcome is that the system remains controllable for realistic friction values. This should be confirmed numerically (e.g. `numpy.linalg.matrix_rank` on $\mathcal{C}$, or `control.ctrb` in the Python `control` package) once actual mass/length/friction values are measured — flagged as a follow-up, not because the answer is expected to change, but because “expected to” is not “verified.”

Part III

Control Design (Phase 4)

8 The LQR Problem

With (A, B) controllable (Part I), the goal is a state-feedback law $u = -K\mathbf{x}$ that stabilizes the upright equilibrium of the *linear* model — this is exactly the Linear Quadratic Regulator problem: choose $u(\cdot)$ to minimize the infinite-horizon quadratic cost

$$J = \int_0^\infty (\mathbf{x}^\top Q \mathbf{x} + u^\top R u) dt, \quad (15)$$

subject to $\dot{\mathbf{x}} = A\mathbf{x} + Bu$, where $Q = Q^\top \succeq 0$ penalizes state deviation and $R = R^\top \succ 0$ penalizes control effort (here R is a scalar since u is scalar, but the theory is stated for the general matrix case since it costs nothing extra).

Tangent — deriving the algebraic Riccati equation from dynamic programming (HJB).

Define the optimal cost-to-go from state $\mathbf{x}$, $V^(\mathbf{x}) = \min_{u(\cdot)} \int_0^\infty (\mathbf{x}^\top Q \mathbf{x} + u^\top R u) dt$ subject to the dynamics starting at $\mathbf{x}(0) = \mathbf{x}$. Bellman’s principle of optimality (splitting the integral into $[0, dt]$ and $[dt, \infty)$ and requiring optimality on each piece) gives the Hamilton–Jacobi–Bellman equation for this time-invariant, infinite-horizon problem:*

$$0 = \min_u \left[\mathbf{x}^\top Q \mathbf{x} + u^\top R u + \nabla V^*(\mathbf{x})^\top (A\mathbf{x} + Bu) \right].$$

For a linear-quadratic problem the optimal cost-to-go is itself quadratic — ansatz $V^*(\mathbf{x}) = \mathbf{x}^\top P \mathbf{x}$ for some symmetric $P \succeq 0$ (this can be verified self-consistently: substituting the ansatz and solving reproduces a valid solution, which is the standard justification), so $\nabla V^*(\mathbf{x}) = 2P\mathbf{x}$ and the bracket becomes

$$\mathbf{x}^\top Q \mathbf{x} + u^\top R u + 2\mathbf{x}^\top P A \mathbf{x} + 2\mathbf{x}^\top P B u.$$

Minimizing over u (unconstrained quadratic in u ; set the gradient w.r.t. u to zero): $2Ru + 2B^\top P \mathbf{x} = 0 \implies$

$$u^* = - \underbrace{R^{-1} B^\top P}_K \mathbf{x}. \quad (16)$$

Substituting u^* back in: using $u^{*\top} R u^* = \mathbf{x}^\top P B R^{-1} B^\top P \mathbf{x}$ and $2\mathbf{x}^\top P B u^* = -2\mathbf{x}^\top P B R^{-1} B^\top P \mathbf{x}$, the bracket collapses to

$$\mathbf{x}^\top \left[Q - P B R^{-1} B^\top P + P A + A^\top P \right] \mathbf{x}$$

(using $2\mathbf{x}^\top P A \mathbf{x} = \mathbf{x}^\top (P A + A^\top P) \mathbf{x}$, valid since a scalar equals its own transpose). The HJB equation requires this bracket to vanish for every $\mathbf{x}$, i.e. the matrix itself must be zero:

$$A^\top P + P A - P B R^{-1} B^\top P + Q = 0. \quad (17)$$

This is the (continuous-time) Algebraic Riccati Equation (ARE). Solving it for the unique $P \succeq 0$ (which exists and is unique given (A, B) stabilizable and $(A, Q^{1/2})$ detectable — both satisfied here, the first by Part I's controllability result, which is strictly stronger than stabilizability) and substituting into (16) gives the LQR gain K .

9 Closed-Loop Stability

Tangent — why the LQR closed loop is guaranteed stable.

This is not automatic for an arbitrary state feedback — it is a specific consequence of P solving the ARE. Consider $V(\mathbf{x}) = \mathbf{x}^\top P \mathbf{x}$ as a candidate Lyapunov function for the closed loop $\dot{\mathbf{x}} = (A - BK)\mathbf{x}$. Its time derivative along trajectories is

$$\dot{V} = \mathbf{x}^\top \left[(A - BK)^\top P + P(A - BK) \right] \mathbf{x}.$$

Since $K = R^{-1} B^\top P$ (symmetric P, R give $K^\top = P B R^{-1}$),

$$K^\top B^\top P = P B R^{-1} B^\top P = P B K,$$

so

$$\begin{aligned} (A - BK)^\top P + P(A - BK) &= (A^\top P + P A) - K^\top B^\top P - P B K \\ &= (P B R^{-1} B^\top P - Q) - P B R^{-1} B^\top P - P B R^{-1} B^\top P \quad [\text{by (17)}] \\ &= -Q - P B R^{-1} B^\top P = -(Q + K^\top R K), \end{aligned}$$

(the last equality using $K^\top R K = P B R^{-1} R R^{-1} B^\top P = P B R^{-1} B^\top P$). Hence

$$\dot{V} = -\mathbf{x}^\top (Q + K^\top R K) \mathbf{x} \leq 0, \quad (18)$$

strictly negative for $\mathbf{x} \neq 0$ whenever $Q \succ 0$ (or, more generally, whenever $(A - BK, Q^{1/2})$ is observable). So $V = \mathbf{x}^\top P \mathbf{x}$ is a strict Lyapunov function for the closed loop — the LQR closed loop is guaranteed asymptotically stable, essentially by construction of the optimal control problem, not as a separate fact requiring extra proof.

10 Solving the Riccati Equation and Choosing Q, R in Practice

Tangent — how `lqr/solve_continuous_are` actually solve (17) numerically.

Equation (17) is a quadratic matrix equation in P — no clean closed form exists for a general 4-state system, so it is solved numerically. The standard method forms the $2n \times 2n$ Hamiltonian matrix

$$H = \begin{bmatrix} A & -BR^{-1}B^\top \\ -Q & -A^\top \end{bmatrix},$$

whose eigenvalues come in $\pm$ pairs symmetric about the imaginary axis (a consequence of H 's special structure). Taking the n eigenvectors associated with the n stable (negative-real-part) eigenvalues and stacking them as $\begin{bmatrix} U_1 \\ U_2 \end{bmatrix}$ (each block $n \times n$), the stabilizing solution is $P = U_2 U_1^{-1}$. This is exactly what `scipy.linalg.solve_continuous_are` and MATLAB's `care/lqr` do under the hood (via a numerically robust Schur decomposition of H , rather than a naive eigendecomposition).

Tangent — Bryson's rule — a starting point for choosing Q, R .

Q, R are design choices, not something derived from the physics. A standard, purely heuristic starting point (Bryson's rule): for diagonal $Q = \text{diag}(q_1, \dots, q_n)$, $R = \text{diag}(r_1, \dots, r_m)$, set

$$q_i = \frac{1}{(\text{max acceptable deviation of } x_i)^2}, \quad r_j = \frac{1}{(\text{max acceptable value of } u_j)^2}.$$

For example, if ± 0.05 rad of pendulum-angle error and ± 0.05 m of cart travel are the largest deviations considered acceptable, and the stepper can comfortably supply up to some $u_{\max}$ before losing steps, those bounds directly seed Q, R — then iterate empirically (increasing q_θ tightens angle regulation at the cost of more aggressive/ noisier control effort, and vice versa for r). This is exactly what was presumably done, at least implicitly, when the existing LQR balancer was tuned by hand — this section formalizes that process rather than replacing it.

11 Applying This to the Hardware

Once the Open TODO physical parameters (m_c, m_p, L, g known exactly, b_x, b_θ to be estimated, e.g. from decay-rate or step-response measurements) are measured, the concrete numeric procedure is:

1. Substitute numeric values into (13) to get numeric A, B .
2. Confirm $\text{rank } \mathcal{C} = 4$ numerically (expected, per Part I, but verify).

3. Choose Q, R (Bryson’s rule as a starting point).
4. Solve (17) numerically for P , then $K = R^{-1}B^\top P$ from (16).
5. Verify $A - BK$ has all eigenvalues with negative real part (guaranteed by §6’s Lyapunov argument, but a cheap direct check).

This K is the same object already running on the Pico; this document’s contribution is deriving *why* it’s the right K from first principles, rather than only having arrived at it by tuning.

12 Summary and What’s Deferred

Established here: the exact nonlinear cart-pole equations of motion (4)–(5) via Lagrangian mechanics (rigid-rod, not point-mass, pendulum); an explicit form (7)–(8) suitable for direct numerical simulation; the linearization (13) about the upright equilibrium; an exact proof of controllability in the frictionless case (14) (with the frictional case argued generic, pending numeric confirmation); and the full LQR derivation, from the optimal control problem (15) through the HJB/Riccati derivation (17), the resulting gain (16), and a first-principles proof of closed-loop stability (18).

Deliberately deferred (per current project scope):

- **Phase 5 — energy-based swing-up.** The energy quantity $E = T + V$ needed for it is already in hand (see the energy tangent in §4), but the swing-up control law itself, capture-region design, and swing-up/LQR hybrid switching logic are not derived in this document.
- **Phase 6 — robustness/polish,** including the idealized- actuator gap noted in §1.1 (real step-per/belt drivetrain dynamics), filtering of the differentiated $\dot{x}, \dot{\theta}$ signals, and travel- limit/safety handling.
- **Numeric parameter measurement** (pendulum mass/length, cart mass, friction coefficients, travel limits) — tracked as an Open TODO in the main project note; every equation above is left symbolic specifically so those values can be substituted in directly once measured.